

11. Implement a Simple Top-Down Parser for Context-Free Grammars

Python Code:

```
grammar = {
    "S": [["NP", "VP"]],
    "NP": [["John"], ["Mary"]],
    "VP": [["runs"], ["eats"]]
}

sentence = input("Enter sentence: ").split()

def parse(symbol, words):
    if symbol not in grammar:
        if words and words[0] == symbol:
            return words[1:]
        return None
    for rule in grammar[symbol]:
        remaining = words
        success = True
        for sym in rule:
            remaining = parse(sym, remaining)
            if remaining is None:
                success = False
                break
        if success:
            return remaining
    return None

result = parse("S", sentence)
if result == []:
    print("Sentence Accepted")
else:
    print("Sentence Rejected")
```

Input

John runs

Output

Sentence Accepted

